Mahi Prashant Nakahte

10

In [ ]:
```
import pandas as pd
import numpy as np
from sklearn import datasets
from sklearn import model_selection
from sklearn import tree

import warnings
warnings.filterwarnings("ignore")
```

In [4]:
```
iris = datasets.load_iris()
print("Dataset structure =", dir(iris))
```

```
Dataset structure = ['DESCR', 'data', 'data_module', 'feature_names', 'filename', 'frame', 'target', 'target_names']
```

In [5]:
```
df = pd.DataFrame(iris.data, columns = iris.feature_names)
df['target'] = iris.target
df['flower_species'] = df.target.apply(lambda x : iris.target_names[x])
```

In [6]:
```
print("Unique target value =",df['target'].unique())
df.sample(10)
```

```
Unique target value = [0 1 2]
```

Out[6]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	flower_species
30	4.8	3.1	1.6	0.2	0	setosa
113	5.7	2.5	5.0	2.0	2	virginica
14	5.8	4.0	1.2	0.2	0	setosa
27	5.2	3.5	1.5	0.2	0	setosa
37	4.9	3.6	1.4	0.1	0	setosa
45	4.8	3.0	1.4	0.3	0	setosa
140	6.7	3.1	5.6	2.4	2	virginica
60	5.0	2.0	3.5	1.0	1	versicolor
114	5.8	2.8	5.1	2.4	2	virginica
119	6.0	2.2	5.0	1.5	2	virginica

In [8]:
```
# Label = 0 (setosa)
df[df.target == 0].head(10)
```

Out[8]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	flower_species
0	5.1	3.5	1.4	0.2	0	setosa
1	4.9	3.0	1.4	0.2	0	setosa
2	4.7	3.2	1.3	0.2	0	setosa
3	4.6	3.1	1.5	0.2	0	setosa
4	5.0	3.6	1.4	0.2	0	setosa
5	5.4	3.9	1.7	0.4	0	setosa
6	4.6	3.4	1.4	0.3	0	setosa
7	5.0	3.4	1.5	0.2	0	setosa
8	4.4	2.9	1.4	0.2	0	setosa
9	4.9	3.1	1.5	0.1	0	setosa

In [9]:
```
# Label = 1 (versicolor)
df[df.target == 1].head(10)
```

Out[9]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	flower_species
50	7.0	3.2	4.7	1.4	1	versicolor
51	6.4	3.2	4.5	1.5	1	versicolor
52	6.9	3.1	4.9	1.5	1	versicolor
53	5.5	2.3	4.0	1.3	1	versicolor
54	6.5	2.8	4.6	1.5	1	versicolor
55	5.7	2.8	4.5	1.3	1	versicolor
56	6.3	3.3	4.7	1.6	1	versicolor
57	4.9	2.4	3.3	1.0	1	versicolor
58	6.6	2.9	4.6	1.3	1	versicolor
59	5.2	2.7	3.9	1.4	1	versicolor

In [12]:
```
# Label = 2 (verginisa)
df[df.target == 2].head(4)
```

Out[12]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	flower_species
100	6.3	3.3	6.0	2.5	2	virginica
101	5.8	2.7	5.1	1.9	2	virginica
102	7.1	3.0	5.9	2.1	2	virginica
103	6.3	2.9	5.6	1.8	2	virginica

In [14]:
```
# Lets create feature matrix X and Y labels
X = df[['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']]
y = df[['target']]

print("X shape=", X.shape)
print("y shape=", y.shape)
```

```
X shape= (150, 4)
y shape= (150, 1)
```

In [16]:
```
X_train,X_test, y_train, y_test = model_selection.train_test_split(X, y, test_size=0.2, random_state = 1)
print("X_train dimenssion =",X_train.shape)
print("X_test dimenssion =",X_test.shape)
print("y_train dimenssion =",y_train.shape)
print("y_test dimenssion =",y_test.shape)
```

```
X_train dimenssion = (120, 4)
X_test dimenssion = (30, 4)
y_train dimenssion = (120, 1)
y_test dimenssion = (30, 1)
```

In [17]:
```
cls = tree.DecisionTreeClassifier(random_state = 1)
cls .fit(X_train,y_train)
```

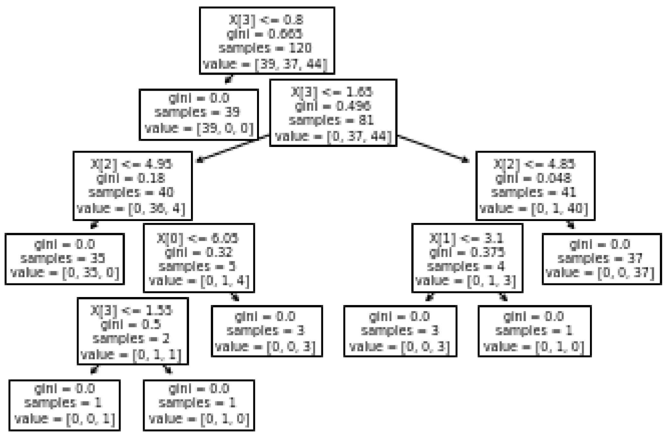
Out[17]:
```
DecisionTreeClassifier(random_state=1)
```

In [18]:
```
cls.score(X_test,y_test)
```

Out[18]:
```
0.9666666666666667
```

In [19]:
```python
tree.plot_tree(cls)
```

Out[19]:
```
[Text(0.4, 0.9166666666666666, 'X[3] <= 0.8\ngini = 0.665\nsamples = 120\nvalue = [39, 37, 44]'),
 Text(0.3, 0.75, 'gini = 0.0\nsamples = 39\nvalue = [39, 0, 0]'),
 Text(0.5, 0.75, 'X[3] <= 1.65\ngini = 0.496\nsamples = 81\nvalue = [0, 37, 44]'),
 Text(0.2, 0.5833333333333334, 'X[2] <= 4.95\ngini = 0.18\nsamples = 40\nvalue = [0, 36, 4]'),
 Text(0.1, 0.4166666666666667, 'gini = 0.0\nsamples = 35\nvalue = [0, 35, 0]'),
 Text(0.3, 0.4166666666666667, 'X[0] <= 6.05\ngini = 0.32\nsamples = 5\nvalue = [0, 1, 4]'),
 Text(0.2, 0.25, 'X[3] <= 1.55\ngini = 0.5\nsamples = 2\nvalue = [0, 1, 1]'),
 Text(0.1, 0.08333333333333333, 'gini = 0.0\nsamples = 1\nvalue = [0, 0, 1]'),
 Text(0.3, 0.08333333333333333, 'gini = 0.0\nsamples = 1\nvalue = [0, 1, 0]'),
 Text(0.4, 0.25, 'gini = 0.0\nsamples = 3\nvalue = [0, 0, 3]'),
 Text(0.8, 0.5833333333333334, 'X[2] <= 4.85\ngini = 0.048\nsamples = 41\nvalue = [0, 1, 40]'),
 Text(0.7, 0.4166666666666667, 'X[1] <= 3.1\ngini = 0.375\nsamples = 4\nvalue = [0, 1, 3]'),
 Text(0.6, 0.25, 'gini = 0.0\nsamples = 3\nvalue = [0, 0, 3]'),
 Text(0.8, 0.25, 'gini = 0.0\nsamples = 1\nvalue = [0, 1, 0]'),
 Text(0.9, 0.4166666666666667, 'gini = 0.0\nsamples = 37\nvalue = [0, 0, 37]')]
```



In [20]:
```python
df.head()
```

Out[20]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target	flower_species
0	5.1	3.5	1.4	0.2	0	setosa
1	4.9	3.0	1.4	0.2	0	setosa
2	4.7	3.2	1.3	0.2	0	setosa
3	4.6	3.1	1.5	0.2	0	setosa
4	5.0	3.6	1.4	0.2	0	setosa

In [21]:
```python
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
sc.fit(X_train)
```

Out[21]:
```
StandardScaler()
```

In [26]:
```python
#input from the user
sepal_length_cm = float(input("Enter new sepal length (cm):"))
sepal_width_cm = float(input("Enter new sepal width (cm):"))
petal_length_cm = float(input("Enter new petal length (cm):"))
petal_width_cm = float(input("Enter new petal width (cm):"))

#prepare the new data point for prediction
new_flower = [[sepal_length_cm, sepal_width_cm, petal_length_cm, petal_width_cm]]

# Transform the data using the sxcaler(assuming 'sc' is already defined)
transformed_new_flower = sc.transform(new_flower)

#pedict the species
result = cls.predict(transformed_new_flower)

#output the result
if result == 0:
    print("The flower specises my be Setosa")
elif result == 1:
    print("The flower specises my be Versicolor")
else:
    print("The flower specises my be Virginica")
```

```
Enter new sepal length (cm):1.2
Enter new sepal width (cm):5.4
Enter new petal length (cm):7.7
Enter new petal width (cm):5.3
The flower specises my be Versicolor
```

In [ ]:

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In [ ]: